

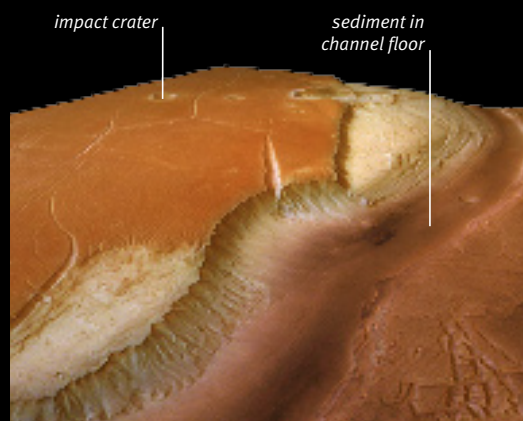
Mars

Mars is the next planet out from the Sun after Earth, and the outermost of the four rocky planets. It is a dry, cold world with a landscape marked by deep canyons and towering volcanoes. Mars is about half the size of Earth, and like Earth it has polar ice caps and seasons, and it spins around in a little over 24 hours.

FEATURES

Mars is made mainly of rock with a small, probably solid iron core. Its rocky surface has been molded by faulting, volcanism, meteorites, water, and wind. Large-scale features such as the Valles Marineris were formed billions of years ago when internal forces split the surface. Elsewhere, huge regions such as the Tharsis Bulge were raised up above the surrounding terrain. The Tharsis Bulge is Mars's main volcanic center and home to giant volcanoes, including Olympus Mons.

Low-lying lava plains cover much of the northern hemisphere. The highland south is older and heavily cratered from meteorite bombardment some 3.9 billion years ago. Dry riverbeds, outflow channels, and floodplains indicate that Mars once had flowing water.



Kasei Vallis

Features such as Kasei Vallis bear witness to the presence of large amounts of fast-flowing water 3–4 billion years ago. This outflow channel, a few hundred miles wide, was probably formed by catastrophic flooding and glacial activity.



Olympus Mons

This is an overhead view of the summit of Olympus Mons. At 15 miles (24 km) high, it is the biggest volcano in the solar system.



Red planet

Much of Mars is rocky, sandy, and dusty, and large areas of its terrain resemble a rock-strewn desert. It is often referred to as the Red Planet, with its coloring coming from iron oxide (rust) in the rocks and soil.

OBSERVATION

Mars is in Earth's sky for much of each year and is one of the easiest planets to see. It is prominent and red and can be spotted with the naked eye by anyone with good eyesight.

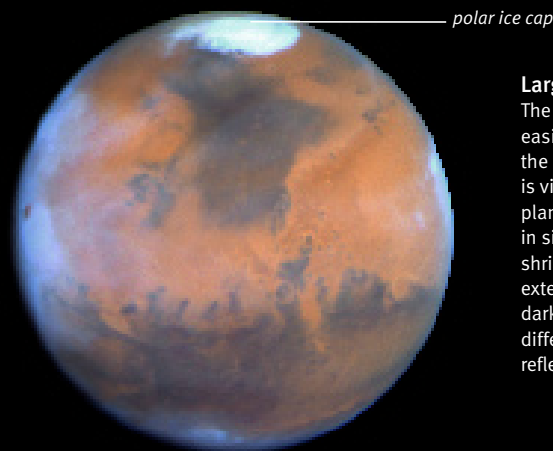
LOCATING AND LOOKING

The best time to see Mars is when the planet is close to Earth and on the opposite side of Earth from the Sun (termed “opposition”; see p.14). This is when it is at its brightest and largest; it is also above the horizon all night long. Oppositions occur roughly every other year, since they are about 26 months apart. Dates of oppositions are included in the Special Events listings in the Monthly Sky Guide (pp.44–69). All oppositions are good for observing Mars, but some are better than others. This is because Mars follows an elliptical orbit around the Sun, and so its distance from us varies. Mars's brightness at opposition is in the range -1.0 to -2.8 .

Mars keeps close to the path of the ecliptic and is found within the zodiac band of sky. It is only some 48.5 million miles (78 million km) away from Earth at opposition and makes rapid progress against the background stars.

It travels west to east but goes into retrograde motion (see p.14) about every 22 months. At such a time, Mars appears to move backward for a few weeks as Earth passes between it and the Sun before once again resuming its forward progression. Mars starts its backward motion about five weeks before opposition.

Mars is the only rocky planet with surface features visible from Earth. Its polar caps can be seen as, in turn, they are tilted toward Earth. Light and dark areas can be seen on the disk; these do not relate to real surface features but result from differences in reflectivity (dark areas reflect the least).



Large-telescope view

The polar ice caps are the easiest features to pick out on the surface of Mars. Just one is visible at a time due to the planet's tilt. The caps change in size with the seasons, shrinking in summer and extending in winter. Light and dark marks (the result of differences in the amount of reflected sunlight) can be seen.



Naked-eye view

Mars is identified by its disk shape and red coloring. Here, it is the brighter of the two dots. Jupiter is to its right.



Binocular view

The planet's disk shape becomes obvious through binoculars, but its surface features are still not visible.



Small telescope view

Here, Mars is visibly red. Some surface features—polar ice caps and dark regions—may be seen.